

DE TOUR VILLAGE, MI, USA

WMO: 997257

Lat: 45.9925N

Lon: 83.8983W

Elev: 180

StdP: 99.19

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-22.2	-19.7	N/A	N/A	N/A	N/A	N/A	N/A	14.9	-5.2	11.9	-5.2	3.9	340	0.621

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.0	26.2	N/A	24.4	N/A	23.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	4.0	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.8	10.2	8.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.7	-8.4	-8.4	-3.4	3.6	9.8	15.6	19.0	18.4	14.7	8.4	2.0	-4.0
	DBStd	10.60	6.39	5.43	5.22	3.77	3.68	3.14	2.81	2.48	3.40	3.70	4.41	5.01
	HDD10.0	2502	569	516	416	194	49	2	0	0	4	76	241	436
	HDD18.3	4713	828	749	674	443	266	92	27	29	114	307	491	694
	CDD10.0	926	0	0	0	1	43	169	278	261	146	27	1	0
	CDD18.3	96	0	0	0	0	1	9	47	32	7	0	0	0
	CDH23.3	270	0	0	0	0	4	33	154	71	8	0	0	0
	CDH26.7	34	0	0	0	0	1	4	22	7	0	0	0	0
(14) Wind	WSAvg	4.0	4.3	4.3	4.4	4.6	4.2	3.9	3.4	3.4	3.5	4.0	4.2	4.3
(15) Precipitation	PrecAvg	803	53	34	42	58	70	81	83	81	89	86	65	57
	PrecMax	1020	121	68	114	109	131	170	185	152	181	171	155	117
	PrecMin	553	16	9	2	16	12	26	32	16	23	30	17	12
	PrecStd	124	25	18	29	25	29	39	37	39	42	36	30	25
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.2	4.1	10.6	16.2	23.7	27.1	29.1	27.6	24.4	18.8	12.4	6.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	3.5	2.0	6.7	12.4	19.9	23.7	26.8	25.2	22.3	16.6	10.6	4.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	2.1	0.8	4.7	10.3	17.4	21.7	25.0	23.7	20.7	15.1	9.0	3.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	0.6	-0.2	3.0	8.5	15.4	20.1	23.4	22.3	19.6	13.8	7.7	2.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	6.8	7.4	7.6	6.6	7.4	7.1	7.0	6.1	5.9	5.0	4.6	5.4
		MCDBR	5.9	7.2	7.9	9.6	10.8	9.4	9.0	7.7	7.0	6.0	5.6	5.6
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.293	0.295	0.321	0.356	0.375	0.396	0.395	0.385	0.366	0.332	0.304	0.285
	taud		2.368	2.362	2.372	2.332	2.371	2.346	2.371	2.397	2.436	2.502	2.515	2.453
	Ebn at Noon		807	885	909	901	892	871	867	863	849	834	798	784
	Edh at Noon		73	90	103	118	118	121	117	110	97	79	64	61
(44) All-Sky Solar Radiation	RadAvg		1.17	2.08	3.52	4.59	5.58	6.03	6.04	5.11	3.91	2.31	1.23	0.87
	RadStd		0.13	0.23	0.35	0.47	0.50	0.46	0.28	0.31	0.33	0.25	0.16	0.09

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page